

STAPHYLOCOCCUS

S. aureus (MSSA, MRSA) and Coagulase-Negative Staphylococci (CoNS)

Microbiology (Bacteriology) • Ananthanarayan & Paniker, Chapter 21

1. INTRODUCTION

- Staphylococci are **Gram-positive cocci in grape-like clusters**; >45 species, many part of normal flora.
- First seen by **von Recklinghausen (1871)**; named by **Sir Alexander Ogston** (Gk *staphyle* = bunch of grapes, *kokkos* = berry). **Rosenbach (1884)** named *S. aureus* for its golden-yellow colonies.
- Two main human pathogens: **S. aureus** (coagulase-positive) and **S. epidermidis** (coagulase-negative). Other CoNS: *S. saprophyticus*, *S. haemolyticus*, *S. lugdunensis*, *S. hominis*.

Test	<i>S. aureus</i>	<i>S. epidermidis</i>
Coagulase	+	–
Mannitol fermentation	+	–
Heat-stable nuclease	+	–
Phenolphthalein phosphatase	+	–
Beta haemolysis on blood agar	+	–
Golden-yellow pigment	+	–
Sensitivity to lysostaphin	+	–

Table 21.1 of the textbook.

2. STAPHYLOCOCCUS AUREUS

Morphology

- Spherical cocci **~1 µm**, in **grape-like clusters** — due to division in **three planes** with daughter cells staying close.
- Non-motile, non-sporing**; a few strains capsulated (young cultures). Uniformly **Gram-positive**; may form **L forms** under penicillin.

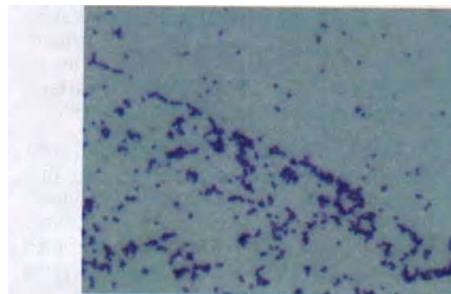


Fig. 21.1 *Staphylococcus aureus* in Gram stain

Cultural Characteristics

- Aerobes and facultative anaerobes**; temperature range 10–42 °C (optimum **37 °C**), pH 7.4–7.6. Grow readily on ordinary media.
- Nutrient agar**: 2–4 mm, circular, convex, smooth, shiny, opaque, easily emulsifiable colonies; **golden-yellow pigment** (best at **22 °C**, aerobic; enhanced by glycerol monoacetate or milk; non-diffusible). Slope shows **"oil-paint" appearance**.
- Blood agar**: most strains **beta-haemolytic** (better under 20–25% CO₂; marked on rabbit/sheep blood). **Sheep blood agar** recommended; human blood should not be used (may contain antibodies/inhibitors).
- MacConkey agar**: small **pink** colonies (lactose fermenter). **Liquid media**: uniform turbidity.
- Selective media**: **salt-milk agar / salt broth (8–10% NaCl)**, **Ludlam's medium** (lithium chloride + tellurite), polymyxin; **Robertson's cooked meat with 10% NaCl** — for faeces, carriers' swabs.

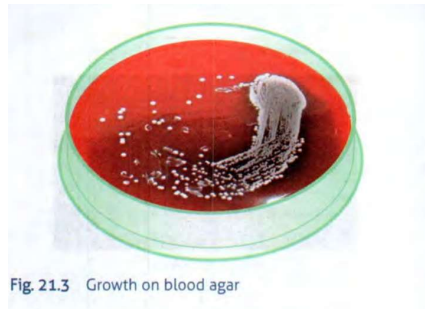


Fig. 21.3 Growth on blood agar

Resistance & Antibiotic Resistance (MSSA vs MRSA)

- Among the **most resistant non-sporing bacteria**. Resistant to **lysozyme** but sensitive to **lysostaphin** (from a strain of *S. epidermidis*).
- **MSSA** (methicillin-sensitive *S. aureus*) = strains sensitive to penicillinase-resistant penicillins (methicillin, cloxacillin).
- **β -lactamase (penicillinase)-mediated resistance**: inducible, **plasmid-mediated** enzyme splitting the β -lactam ring; transmitted by **transduction or conjugation**. Such strains remain sensitive to **methicillin, cloxacillin** and **β -lactam + β -lactamase inhibitor** combinations (amoxicillin-clavulanic acid).
- **Altered target site — PBP2a**: altered penicillin-binding protein reduces affinity for **all β -lactams** → **MRSA**. Encoded by **mecA gene** on the chromosomal **SCCmec** (staphylococcal cassette chromosome mec). Usually also resistant to erythromycin, tetracycline, aminoglycosides; cause hospital outbreaks.

Feature	HA-MRSA	CA-MRSA
SCCmec type	I, II, III	IV
Resistance	Multidrug-resistant	Less resistant
PVL toxin	Less common	More likely to produce PVL
Transmissibility	—	More transmissible

- **EMRSA** (epidemic MRSA) spread easily in hospitals → post-operative wound and cross-infections. The division between HA and CA-MRSA is now blurring.
- **Tolerance**: bacterium only **inhibited, not killed** by penicillin — wide gap between **MIC** and **MBC**.
- **VRSA**: vancomycin-resistant; harbour both **vanA** and **mecA** genes; susceptible only to **linezolid**. **VISA**: intermediate; **no resistance genes but thickened cell wall** (prolonged vancomycin therapy). **hVISA** = heterogeneous VISA.

MEMORY AID

- MRSA = **mecA** → **PBP2a** → **all β -lactams fail**.
- CA-MRSA = **SCCmec IV + PVL** ("Community **IV** drip of **PVL**").
- **VISA = thick wall, no gene**; **VRSA = van gene**.

Virulence Factors

- Staphylococci cause two types of disease: **infections** (invasion, multiplication) and **intoxications** (preformed toxin).

Cell-associated factors

- **Peptidoglycan** — rigidity; activates complement, induces cytokines. **Teichoic acid** — adhesion; protects from complement-mediated opsonisation. **Capsule** — inhibits opsonisation.
- **Protein A**: binds **Fc of IgG** (except **IgG3**), leaving Fab free → basis of **co-agglutination**. Antiphagocytic, anticomplementary, chemotactic; B-cell mitogen; used to isolate IgG.
- **Clumping factor (bound coagulase)** → slide coagulase test. Receptors for **fibronectin, fibrinogen, IgG, C1q** → adhesion.

Extracellular enzymes

- **Coagulase (free)**: clots human/rabbit plasma with **coagulase-reacting factor (CRF)**; fibrinogen → fibrin. Basis of **tube coagulase test**.
- **Lipases, hyaluronidase** (breaks connective tissue), **staphylokinase (fibrinolysin)**, proteases. **Heat-stable DNase (thermonuclease)** — characteristic of *S. aureus*.

Toxins

Toxin	Key features
Alpha haemolysin	Most important; lyses rabbit RBCs ; inactivated at 70 °C but reactivated at 100 °C (paradoxical). Leucocidal, cytotoxic, dermonecrotic, neurotoxic, lethal
Beta haemolysin	Sphingomyelinase; lyses sheep RBCs ; hot-cold phenomenon (lysis starts at 37 °C, seen on chilling)
Gamma haemolysin	Two separate proteins, both needed
Delta haemolysin	Detergent-like action on RBCs, leucocytes, macrophages, platelets
Panton-Valentine leucocidin (PVL)	Two-component (S & F) synergohymenotropic toxin; associated with CA-MRSA
Enterotoxin	8 types (A, B, C1–3, D, E, H); type A most common. Heat-stable (100 °C for 10–40 min). Acts on autonomic nervous system → food poisoning. Superantigen
TSST-1	Toxic shock syndrome toxin (formerly enterotoxin F / pyrogenic exotoxin C); phage group I strains; superantigen
Exfoliative toxin (exfoliatin)	Staphylococcal scalded skin syndrome (SSSS) — Ritter's disease (newborn), toxic epidermal necrolysis; milder: pemphigus neonatorum, bullous impetigo

Staphylococcal Diseases

- Characteristically **localised pyogenic lesions** (vs spreading streptococcal lesions).
- Skin & soft tissue**: folliculitis, **furuncle (boil)**, abscess (breast abscess), wound infection, **carbuncle**, impetigo, paronychia, cellulitis (less often).
- Musculoskeletal**: osteomyelitis, arthritis, bursitis, pyomyositis. **Respiratory**: tonsillitis, pharyngitis, sinusitis, otitis, bronchopneumonia, lung abscess, empyema.
- CNS**: abscess, meningitis, intracranial thrombophlebitis. **Endovascular**: bacteraemia, septicaemia, pyaemia, **endocarditis**. **Urinary**: with instrumentation, implants or diabetes — significant even at low counts.
- Toxin-mediated**: **food poisoning** (nausea, vomiting, diarrhoea **2–6 h** after food; no fever; self-limited; source = food handler carrier; milk products, meat, fish), **toxic shock syndrome** (fever, hypotension, rash that desquamates; tampon-related, now rare), **SSSS**.

Epidemiology & Hospital Infection

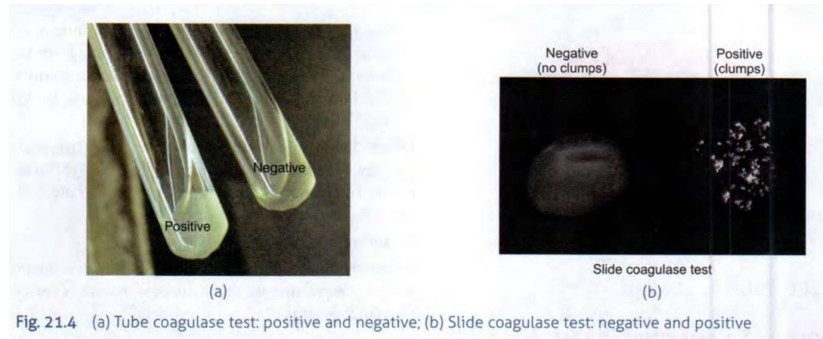
- Source: human patients and **carriers**. Carriage: **nose 10–30%**, perineum ~10%, vagina 5–10% (rises during menses). **Shedders** disseminate large numbers for long periods.
- Transmission by contact (direct / fomites), dust, airborne droplets. Colonisation of **umbilical stump** common in hospital-born babies.
- Control**: standard precautions — **hand washing** (simplest, most effective), isolation of patients with open lesions, keeping infected staff off work, asepsis in theatres. Nasal carriers: **mupirocin**; chlorhexidine; rifampicin + another oral drug for resistant cases.
- Typing**: phage typing (phenotypic; now obsolete — MRSA mostly phage group III), antibiograms; **molecular** — **PFGE**, **MLST**, **RFLP**, ribotyping (more discriminatory).

Laboratory Diagnosis

- Specimen**: pus, sputum; faeces & remains of food (food poisoning); **nasal swab** (carriers); perineum, hair, umbilical stump.
- Microscopy**: Gram-positive cocci in clusters (useful for pus; not for sputum with mixed flora).
- Culture**: blood agar (overnight); selective media (salt-milk agar, Ludlam's, cooked meat with 10% NaCl) for mixed samples.
- Catalase positive** (unlike streptococci); hydrolyse urea, reduce nitrate, liquefy gelatin, MR & VP positive, indole negative.

	Tube coagulase	Slide coagulase
Detects	Free coagulase	Bound coagulase (clumping factor)
Mechanism	Activates CRF → coagulase-CRF complex acts on fibrinogen → clot	Reacts directly with fibrinogen → cells clump; no CRF needed
Method	0.1 ml culture + 0.5 ml human/rabbit plasma; 37 °C water bath, 3–6 h ; clot does not flow on tilting	Emulsify in saline, check no autoagglutination, add plasma → prompt clumping

	Tube coagulase	Slide coagulase
Note	EDTA, oxalate or heparin plasma — not citrate (false positive); don't incubate longer (fibrinolysin lyses clot)	Some CoNS may be positive → confirm by tube test



- **MRSA detection: cefoxitin disc diffusion** (correlates with *mecA*). **BORSA** (borderline resistant) may be missed → **oxacillin screen agar**.
- **D-test**: clindamycin disc near erythromycin disc; flattening (D-shaped zone) = **inducible clindamycin resistance** → report clindamycin resistant.
- **Vancomycin**: test by **MIC only** (not disc diffusion). Serology — limited value; molecular methods for outbreaks.

Treatment

- Choose drug on **sensitivity testing**. **Benzyl penicillin** — most effective if non-penicillinase producer. **Cloxacillin** — penicillinase producers (MSSA).
- **Vancomycin / teicoplanin** — drug of choice for **MRSA**. **Linezolid** — for **VRSA**. Mild superficial lesions: topical **bacitracin, chlorhexidine, mupirocin**.

CLINICAL CORRELATION

- Food poisoning (short incubation 2–6 h, vomiting, no fever) — preformed **heat-stable enterotoxin**; reheating food does not help.
- Nasal carriage in food handlers and hospital staff is the key source of outbreaks.
- CA-MRSA with **PVL** → severe skin infections and necrotising pneumonia.
- Don't report clindamycin sensitive if **D-test positive**.

3. COAGULASE-NEGATIVE STAPHYLOCOCCI (CoNS)

- Major part of **normal skin flora**; pathogenic species: *S. epidermidis*, *S. haemolyticus*, *S. lugdunensis*, *S. saprophyticus* — mostly **device-associated** and **healthcare-associated** infections.
- ***S. epidermidis***: most common CoNS pathogen; invariably on skin. Causes **stitch abscess**; infects **implanted foreign bodies** — prosthetic valves, shunts, **IV catheters** → bacteraemia, **central-line-associated BSI**, cystitis, endocarditis (drug addicts). Hospital strains usually multidrug-resistant.
- **Biofilm** (extracellular polysaccharide matrix) — key virulence factor; protects from antibiotics and aids colonisation.
- ***S. saprophyticus***: skin and periurethral area; **UTI in sexually active young women**; sensitive to most antibiotics except nalidixic acid; identified by **novobiocin resistance** (*S. epidermidis* is novobiocin sensitive).
- **Other coagulase-positive staphylococci**: *S. intermedius*, *S. hyicus* — animal parasites, do not infect humans.
- **Micrococci**: Gram-positive cocci, larger, in **pairs, tetrads** or irregular clusters; catalase and **modified oxidase positive**, **oxidative** in Hugh-Leifson test; usually non-pathogenic.

★ GOLDEN ONE-LINERS

- Staphylococci: **catalase positive**; streptococci: catalase negative.
- Coagulase test separates **S. aureus** from CoNS; tube test = free coagulase, slide = clumping factor.
- Protein A binds Fc of IgG (not **IgG3**) → **co-agglutination**.
- Alpha toxin lyses **rabbit** RBCs; beta toxin lyses **sheep** RBCs with **hot-cold phenomenon**.
- Enterotoxin **type A** commonest; toxin resists **100 °C**.
- MRSA: **mecA** on **SCCmec**, encodes **PBP2a**; detected by **cefoxitin disc**.
- *S. saprophyticus* = **novobiocin resistant**; *S. epidermidis* = **biofilm** on devices.

🎯 EXAM FOCUS

- Virulence factors of *S. aureus* with mechanisms — essay favourite; make a table of toxins.
- MRSA: mechanism (*mecA*/PBP2a), HA vs CA-MRSA, detection (cefoxitin), treatment (vancomycin).
- Coagulase test: tube vs slide, principle and interpretation.
- Differences between *S. aureus* and *S. epidermidis* (Table 21.1).

📄 EXAM QUESTIONS (from the textbook)

Essays

1. Describe the virulence factors of *Staphylococcus aureus* and their mechanism of action.
2. What is toxic shock syndrome? Describe its etiology and pathogenesis.

Short answers / Short notes

1. Co-agglutination (definition)
2. Differences between *Staphylococcus aureus* and *Staphylococcus epidermidis*
3. Principle and interpretation of the coagulase test
4. MRSA; Staphylococcal food poisoning